LAB 2: INTERACTIVE PORTFOLIO ENHANCEMENTS

Course DA558A HT26 JavaScript for Web Development (Grade 5)

Report and Reflection

1. Project Information

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Target Grade: 5

Repository: https://github.com/tinaob/portfolio-lab2

Project overview

For Lab 2, I did extend my Lab 1 portfolio with features that are interactive, which uses HTML, CSS and vanilla JavaScript. The work does includes a feedback form with validation as well as user feedback, a skills showcase that is generated from a JavaScript array, and also an accessibility display panel that allows visitors to be able to change the text size and also contrast. I did test the features in the browser, and then I used the developer tools to check the behaviour, and also made changes when the results did not match what I expected.

Features implemented

- Feedback form: This demostrates the name, email, topic, comments as well as website field indicating ‘optional’, and then it displays field-level errors and success feedback which includes a live comments counter as well as Clear button.
- Data-driven skills showcase: This renders eight skills from an array, filters by category, sorts by proficiency, years of experience or name (A-Z), and then also updates the counter of the visible-results.
- Accessibility display panel: This provides text settings that is Small, Medium and Large plus a Contrast control that is Normal/High which uses visible keyboard focus as well as pressed states.
- Persistent settings: this saves the display settings that is selected so that they remain in effect after a reload as well as across the pages of the portfolio.
- Accessibility and motion checks: The keyboard operation was tested, the readability of high-contrast as well as the reduced-motion CSS behaviour.

2. Feature Evidence

2.1 Feedback form-normal state

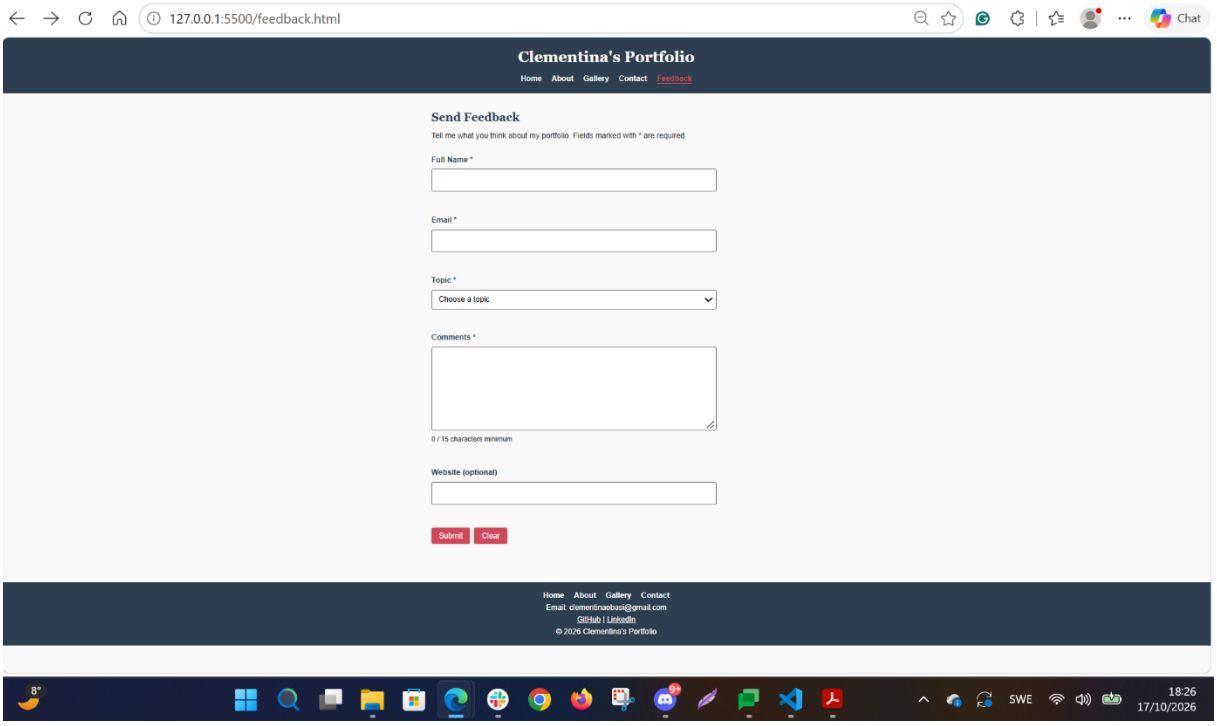


Figure 1. This is the feedback form when it is in its initial state, which shows that the fields are required, the Website field that indicates “optional”, the buttons for Submit and Clear, as well as the counter for comments.

2.1 Feedback form-validation errors

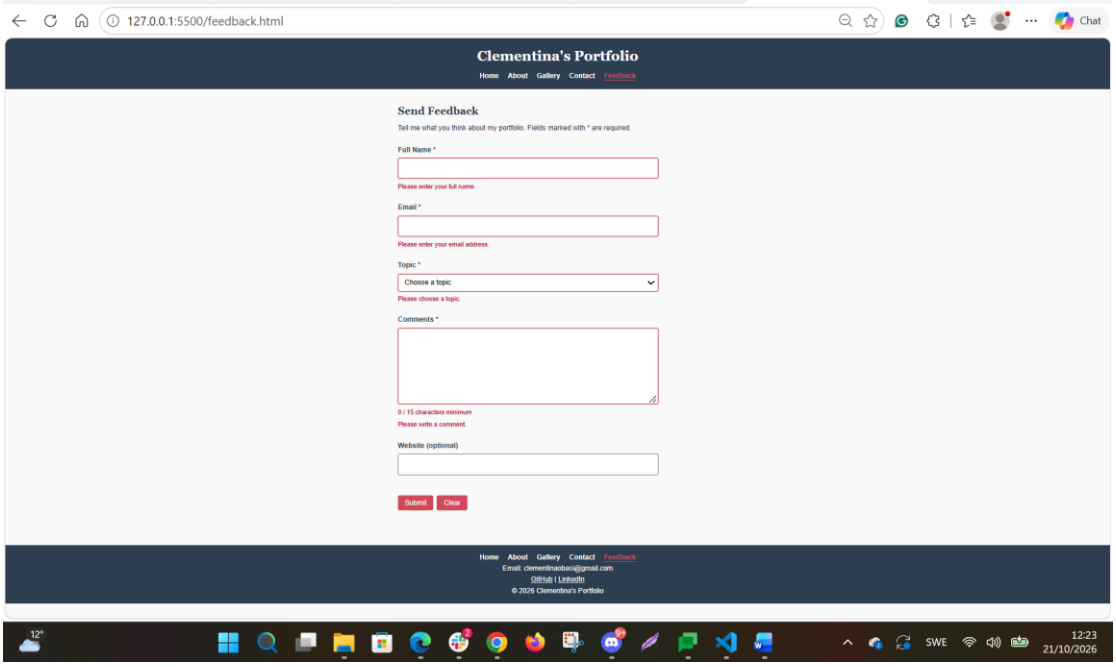


Figure 2. When I submitted an empty form, errors showed beside the fields that are required. When the field for the optional Website is empty it does not show any error.

2.1 Feedback form-valid input and counter

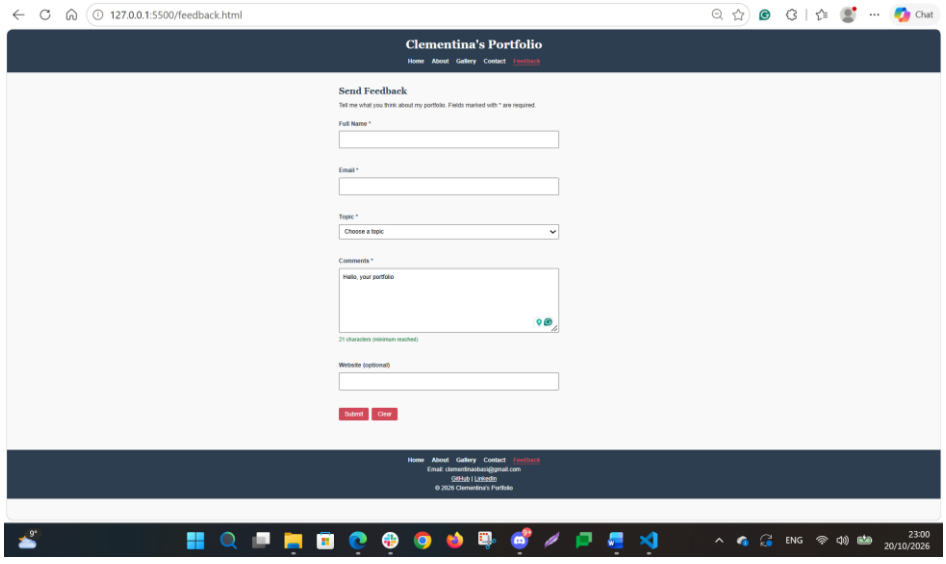


Figure 3. When the styling is successful, then the valid input is shown. The counter for the comments changes as text is entered and then it indicates when the 15-character minimum has gotten to the limit or reached.

2.1 Feedback form-submission was successful

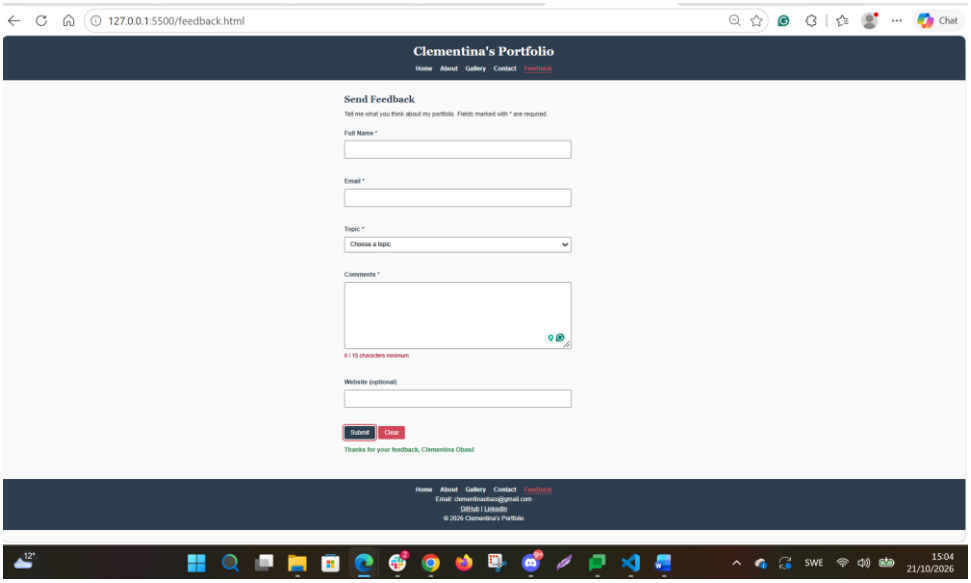


Figure 4. After successful and valid submission, the page showed a thank-you message that is personalised, and then it resets the form as well as removes the message after approximately four seconds.

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2.2 Skills-all skills

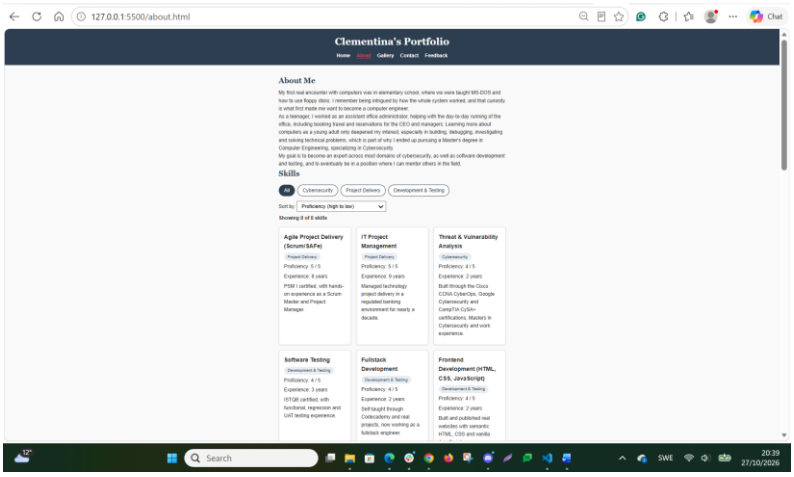


Figure 5. The skills section showed all the eight cards as well as the counter with the results that are visible.

2.2 Skills-filtered and sorted

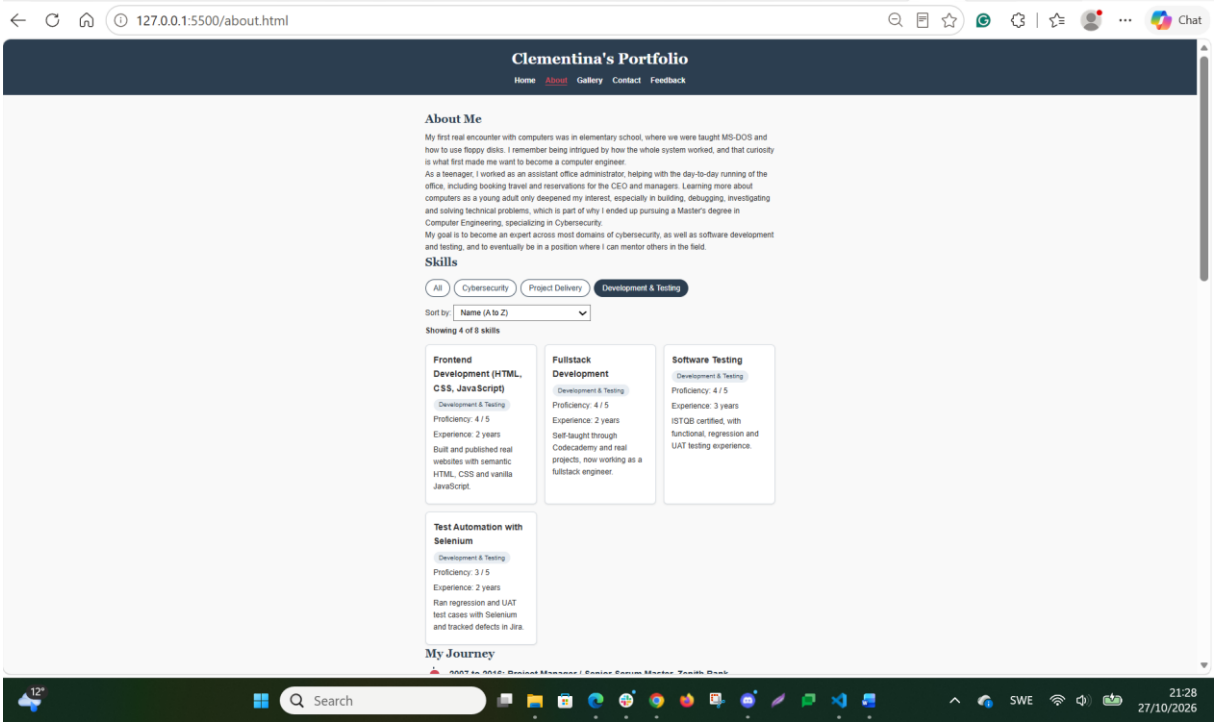


Figure 6. A filter for category and a sort option are applied together here. The cards that are visible remain in the category that is selected as well as are ordered by the chosen sort option.

2.3 Display panel-default settings

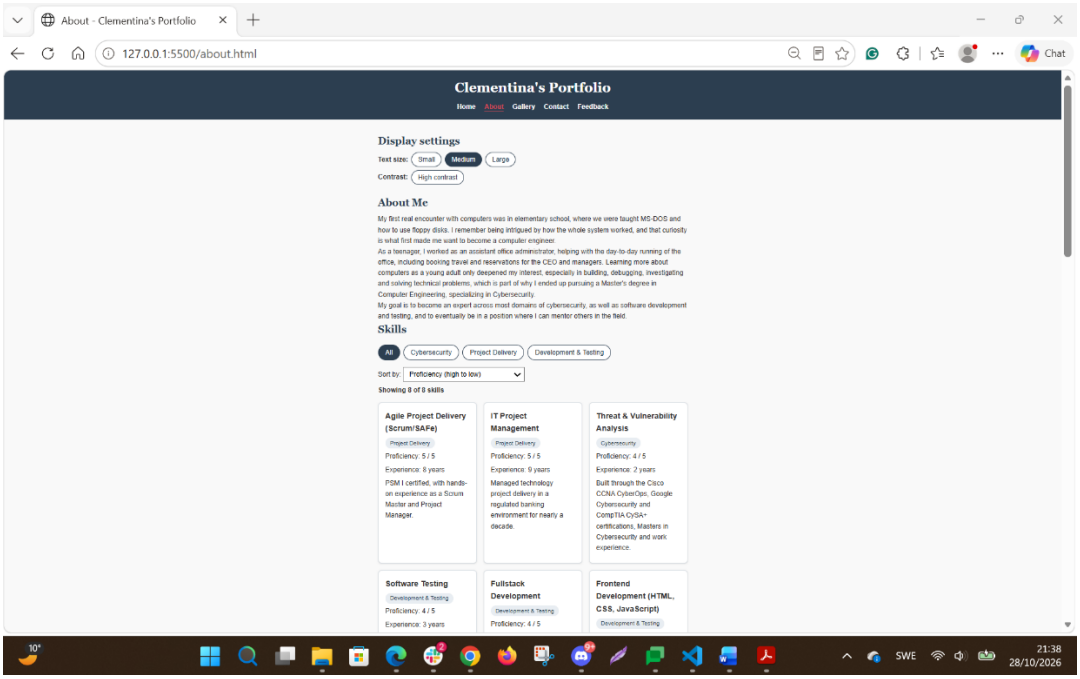


Figure 7. I started by building display panel that starts with Medium text as well as Normal contrast.

2.3 Display panel-Large text and High Contrast

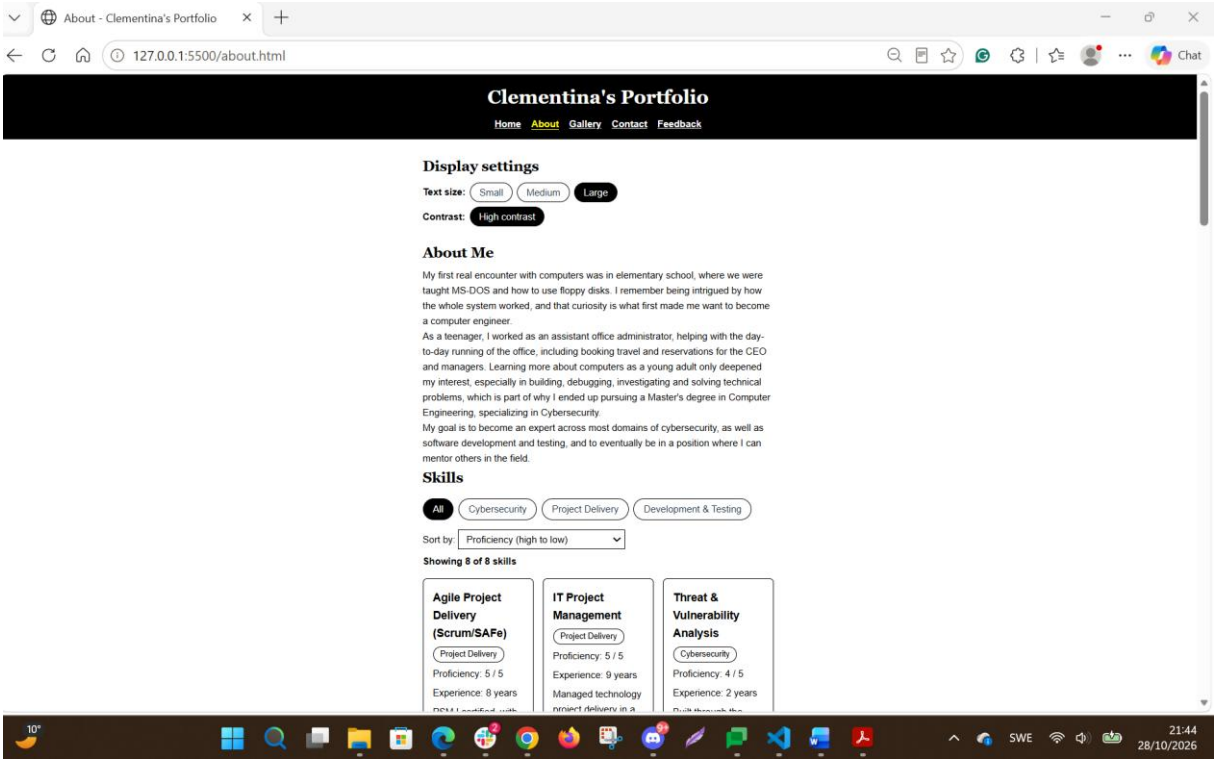


Figure 8. I then built and enabled Large text and High Contrast. The contrast , borders, links as well as focus indicators that are stronger, make the interface elements easier to differentiat.

2.3 Display panel-persistence after reload

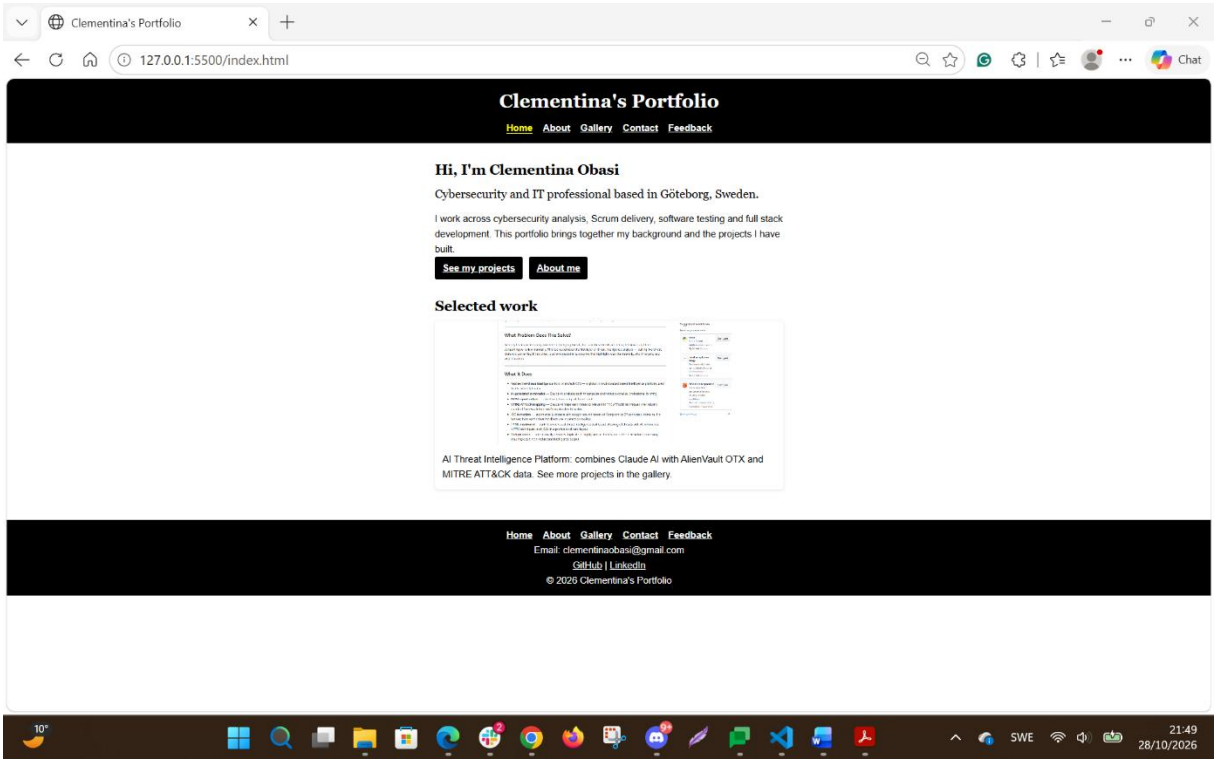


Figure 9. After I have reloaded or navigated to another portfolio page, the text size that i previously selected as well as contrast settings are restored.

3. Code Explanation

3.1 Feedback Form and Validation

Feedback form utilizes validation functions that are separate so as to enforce different rule for each individual field. validateFullName() further checks to ensure a name is present, this has minimum of 2 characters and utilizes characters that are allowed. validateEmail() also checks to make sure the email format is met, while validateTopic() checks to make sure topic is selected by visitor. validateComments() checks to make sure comment fields are not left blank, and also meet the 15-character minimum. The optional Website field is only when it contains a value, which is achieved when a visitor inputs a web address, without the right format, the field further adds whats missing

showFieldError() further shows error message and applies error styling to the relevant field. clearFieldError() further removes both error styling and message when field becomes valid. As visitors changes and types feedback when minimum length is reached, the comments counter automatically updates. Upon submission, event.preventDefault() further prevents page from reloading while JavaScript also validates the form. Once a required fields pass, a personalised thank-you message appears, and triggers the reset of resetForm(), further returning it to its initial state, while setTimeout() further removes message after about four seconds. The form does not store feedback on a server and does not send an email, however, it is browser-only

3.2 Skills cards, filtering and sorting

The skills that were showed uses an array of eight objects. Each of those object has an id, name, category, proficiency, yearsExperience as well as description. The renderSkills() function does create the cards that are visible from this data, so the cards does not need to be written in an individual way in the HTML. The filter buttons does allow visitors to show All skills or select Cybersecurity, Project Delivery, or Development & Testing. The dropdown for sort, orders the visible skills by proficiency, years of experience, or the name from A to Z. The logic for filtering and sorting works together by filtering the data first, sorting a copy of the results that have been filtered, rendering the cards as well as updating the counter for “Showing X of Y skills”.

The buttons for filter use data-category to know which of the category they represent. The aria-pressed state communicates which of the filter is active to technology that has accessible tools, while the dark fill gives a visible indication of the button that is active. When I pressed Tab, a red outline showed keyboard focus. This is different from the dark fill, which shows the filter that is active.

3.3 Accessibility display panel and persistence

The display panel offers High Contrast toggle and a combination of Small, Medium and Large text settings, the applySettings() function further applies selected settings through updating controls’ active/pressed states, and page classes. Base text sizes in this case are either 14px, 16px or 20px, background and foreground colours are changed by high contrast, further strengthens the borders and keeps the links as well as focus indicators visible. localStorage is responsible for where the setting is saved, and can be restored when page loads, further allowing selected displays remain consistent, even after reloading and moving between pages of the portfolio

3.4 Tracing an interaction: changing text size

When a text size button is activated by a visitor, identification of requested size takes place through event handler. Additionally, selected setting is also updated while applySettings() further applies matching class to the page, and also update active button state. This is eventually saved in localStorage, when page loads again, saved value is red and applied at the same time, and this make sure visitor does not select same size again.

4. Problem I Solved

Problem 1: Ease of Use of optional Website field

What I notice is that there is a possibility of visitors to type familiar address like www.tina.com without knowing that a URL automatically needs a protocol such as https://. I wanted a situation where the form to be flexible, instead of them having to understand the technical detail. This made me implement validation for the optional Website field and further added https://, so that when visitor entered a valid address without a protocol, it would still go through. However, I also tested an invalid value like “tina”, this showed an error, then I further tested “www.tina.com”, this was instantly processed and displayed the URL as “https://www.tina.com”. The lesson learnt from this is that both user and technical validations perspective matter a lot.

Problem 2: Keeping navigation readable in High Contrast mode

When High Contrast was enabled, there was the appearance of active navigation link which was dark red in colour, as against black header which made it very difficult to read. This issue came from the active and hover colours that were used, which is accent colour in the high-contrast styling. The navigation link colour was eventually changed to yellow, from previous High Contrast mode, this made it clearly visible against the black background. When I finally checked navigation at the top and bottom of the pages, it showed the yellow links became easier to be identified and distinguished.

5. Accessibility and Motion Testing

The display panel was tested using only the keyboard, this was done by clicking on empty part of the page and further pressing tab first, then text-size button received focus, which continued through high contrast, as well as Small, Medium, Large text. Enter button was also used in activating Large and Space to toggle High Contrast, while Shift+Tab was used to move backwards. All the four control proved to be reachable, and furthermore, the red focus outline maintained visibility, while both Space and Enter worked as well.

Additionally, display panel's six combinations was personally tested, which include Small, Medium and Large text with both High and Normal contrast. In addition, pages after reloading and changing settings were also check, selected contrast and text size remained applied across pages of portfolio. Browser Console showed no JavaScript errors during check that were recorded in my notes. Furthermore, reduction in motion in browser rendering panel was also checked, especially when prefers-reduced-motion set to reduce. Card did not hover, while filter colour changed without fading. However, when reduced-motion emulation is turned off, there was smooth lifting of the card. One lesson that I learnt is to repeat a test in the normal browser view, especially if device toolbar has the potential to affect the result.

6. Reflection

6.1 Which interaction am I most confident explaining, and why?

Where I had the most confidence was when explaining the sort interaction and skills filter, this is because it was built and tested in small steps. Arrays acts as storage for the skills, page create card from data, and when any category is selected, the code filters the skills. In addition, when a sort option is chosen, sorting is done on results that are filtered and further renders the updated cards, furthermore, counter changes to show how many skills are available. This made me to also understand red outline and dark button, the former shows keyboard focus while latter shows active filter. Testing different combination further assisted me in being able to see how the logic of JavaScript affects what takes place on the page

6.2 How does my Git history show the solution changing?

The entirety of my work was done in stages instead of been completed in one step, this started with both feedback page structure and portfolio baseline. This then connected JavaScript and added the form validation one field at a time. Once the testing was complete, skills array and rendering, filtering and sorting were also added, while I continued to work on accessibility styling and display panel. The changes demonstrated a process which include implementation, testing, identifying any issue and problem, and then eventually refining the results.

- **Website field:** b42119e (2026-10-21), then e0999b5 (2026-10-21). The first one checks the field, and then the second one fixes when people typ a website without https://.

- **Skills:** 2343065 (2026-10-26), then 2f7b119 (2026-10-27). Filtering started on its own, and then was combined with sorting as well as a counter.
- **Display panel:** 58786dc (2026-10-28), then f1f50ec (2026-10-28). Settings were first only remembered, and then applied on all pages.

6.3 With one more development session, what would I improve?

Having gone through another session development, these are the things I will do

1. Focus on both testing and refining the experience, especially across different screen resolutions with different user needs
2. Check that mobile user find it easy to complete feedback form, and that both category labels and long skill descriptions does not make card uneven, and display panel remains readable, especially at Large text in High Contrast mode.
3. Review all captions for screenshots, just to make sure they all show actual final behaviour, and further verify that every page saves and restores setting differently
4. Finally review my report and Git history to further make sure evidence is clear, and that explanations can be given to relevant functions confidently, especially the seminar.